Summary in Graph

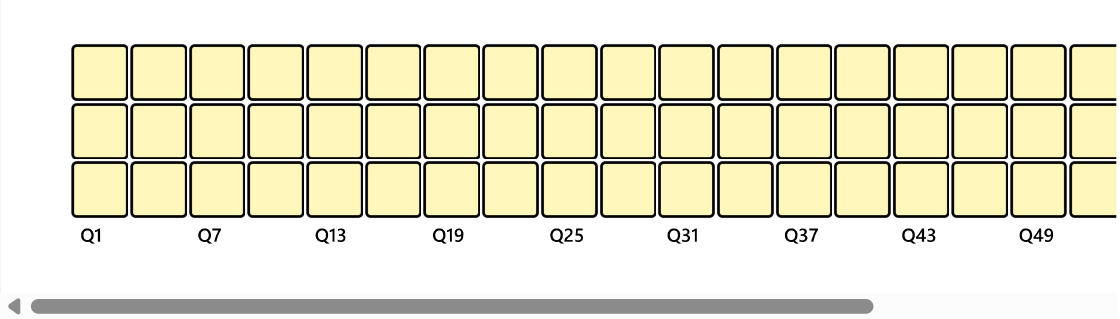
Exam Summary (GO Classes DA Test Series 2026 | MOCK GATE | Test 4)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	65 (30 + 35)
Total Marks:	100 (30 + 70)
Exam Duration:	180 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (20) out of 198 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Aptitude

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Analytical Aptitude

The sequence of the five symbols ◯ ◄ ☒ △ ★ repeats to form the pattern:

◯ ◄ ☒ △ ★ ◯ ◄ ☒ △ ★ ⋯

If the pattern is continued, the 23rd symbol in the pattern is

A. 

B. 

C. 

D. 

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

The average (mean) of a list of 10 numbers is 17. When one number is removed from the list, the new average is 16. What number was removed?

A. 26
B. 36
C. 17
D. 16

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 07sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

When a positive integer n is divided by 10, the remainder is 8.
When n is divided by 5, the remainder is _____.

A. 0
B. 1
C. 2
D. 3

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

An integer is randomly chosen from the list $10, 11, 12, 13, 14, 15, 16, 17, 18, 19$. What is the probability that the chosen integer is even?

A. $\frac{3}{10}$

B. $\frac{4}{10}$

C. $\frac{5}{10}$

D. $\frac{6}{10}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

Choose the most appropriate word from the options given below to complete the following sentence:
Given the seriousness of the situation that he had to face, his ___ was impressive.

A. beggary
B. nomenclature
C. jealousy
D. nonchalance

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

Glen, Hao, Ioana, Julia, Karla, and Levi participated in the 2023 Canadian Team Mathematics Contest. On their team uniforms, each had a different number chosen from the list $11, 12, 13, 14, 15, 16$. Hao's and Julia's numbers were even. Karla's and Levi's numbers were prime numbers. Glen's number was a perfect square. What was Ioana's number?

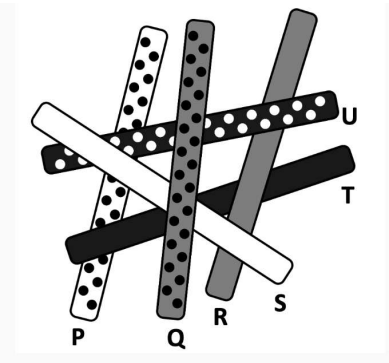
A. 11
B. 15
C. 14
D. 13

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

Ana drops six sticks on a table as shown.
Then she picks all the sticks up according to the following rules:

1. Pick up one stick at a time.
2. Only pick up a stick if no other stick is on top of it.



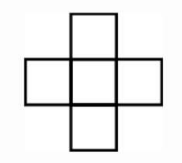
In which order did Ana pick up the sticks?

A. S, Q, U, T, R, P
B. T, Q, P, U, S, R
C. Q, S, U, R, T, P
D. Q, S, T, U, P, R

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

Five different integers are selected from 1 to 6 and one integer is placed into each of the five squares shown.



The integers are placed so that the sum of the three integers in the vertical column is 7, and the sum of the three integers in the horizontal row is 11.
Which integer does not appear in any square?

A. 3
B. 4
C. 2
D. 5

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

A container of ice cream can make 6 cones or it can make 4 sundaes. If 5 such containers of ice cream are used to make 12 cones, what is the greatest number of sundaes that can be made with the ice cream that remains?

A. 4
B. 8
C. 12
D. 16

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Verbal Aptitude

Although Mounik admits that the _____ of his career is to make Blue Bunny a house name, he is not _____ about achieving his goals.

A. worst part, concerned
B. unspoken aim, bothered
C. biggest challenge, unduly worried
D. ultimate goal, prepared

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Artificial Intelligence

Which of the following statements correctly defines the Markov Blanket of a Node in a graph? NOTE : $X \perp Y$ represents X is independent of Y and $(X \perp Y \mid Z)$ represents that X is independent of Y given a set of variables Z. The symbol "/" denotes set difference.

A. The Markov Blanket of a node x in a graph with verte x set X is the smallest set Z such that $x \perp X/\{Z \cup x\} \mid Z$.
B. The Markov Blanket of a node x in a graph with vertex set X is the largest set Z such that $x \perp X\{Z \cup x\} \mid Z$.
C. The Markov Blanket of a node x in a graph with vertex set X is the smallest set Z such that $\mathrm{x} \perp \mathrm{X}/\{\mathrm{x}\} \mid \mathrm{Z}$.
D. The Markov Blanket of a node x in a graph with vertex set X is the largest set Z such that $x \perp X/\{x\} \mid Z$.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Select Type | Award: 1 | Penalty: 0 | Artificial Intelligence

Suppose we try to get driving directions from Chicago to New York City using A* Algorithm. Which of the follwing can A^* return? Assume nothing about the heuristic function.

A. A correct shortest paths tree to every city from Chicago.
B. A incorrect shortest paths tree to every city from Chicago. That is, every such path is suboptimal.
C. A correct shortest path from Chicago to New York.
D. An incorrect shortest path from Chicago to New York.

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Artificial Intelligence

Which of the following formulas is a formalization of the sentence:
"Not all cars have carburetors."

A. $\neg\forall \mathrm{x}[\mathrm{car}(\mathrm{x}) \rightarrow \text{carburetors } (\mathrm{x})]$
B. $\forall \mathrm{x}[\mathrm{car}(\mathrm{x}) \rightarrow \text{carburetors } (\mathrm{x})]$
C. $\exists \mathrm{x}[\mathrm{car}(\mathrm{x}) \wedge \neg \text{ carburetors } (\mathrm{x})]$
D. $\neg\exists \mathrm{x}[\mathrm{car}(\mathrm{x}) \wedge \neg \text{ carburetors } (\mathrm{x})]$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

We trained a multiclass logistic regression model with the following weight matrix:

$$\theta = [\theta_A \theta_B \theta_C] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Here, θ_A, θ_B, and θ_C are the weight vectors for classes A, B, and C , respectively. Let the data point $x = [1, -1, 1]^T$. Determine the class label assigned to x based on the softmax output.

A. Class A
B. Class B
C. Class C
D. There is a tie, and the model cannot decide

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in Python

Given the string s = "GATE DA 2025", what is the output of the following expression?

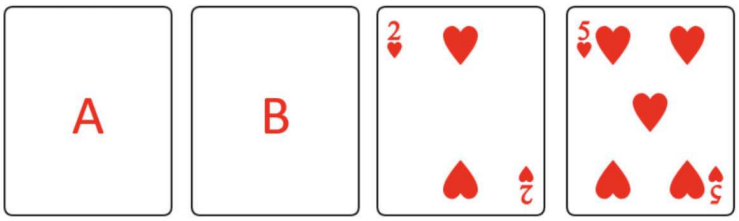
```
s[:-1]
```

A. 'GATE DA 202'
B. 'GATE DA 2025'
C. '202 AD ETAG'
D. ' 5202 AD ETAG'

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Numerical Type | Award: 1 | Penalty: 0 | Artificial Intelligence

Each card has a letter on one side and a natural number on the other side. How many cards do you have to turn over to determine whether the following claim is true?
Claim: For each card, if the card has a vowel on one side, then it has an even number on the other side.



Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Which of the following is/are TRUE?

A. In the worst case, merge sort runs in $O\left(n^2\right)$ time.

B. Depth-first search of a graph is asymptotically faster than breadth-first search.

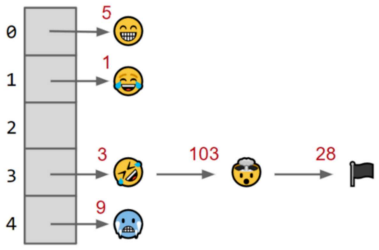
C. Dijkstra's algorithm is an example of a greedy algorithm.

D. Insertion in a binary search tree is "commutative". That is, inserting x and then y into a binary search tree leaves the same tree as inserting y and then x.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider a hashtable, where we represent each bucket with a linked list.
Suppose we have added the Picture objects below with the given hashcodes in red:



We add a new picture 👽 with hashcode **-6**. In which bucket will 👽 end up in the hashtable?

A. 0
B. 2
C. 3
D. 4

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

After applying the following operations to an empty stack:

push(35), push(36), push(43), push(8), pop, pop, push(51), pop

what is the content of the stack?

A. 35,36
B. 36,43
C. 35,8
D. 35,51

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Provided the full SVD of a square full-rank matrix X is written as $X = U\Sigma V^T$, which of the following matrixes are always symmetric? Select all that apply.

A. $U^T X$
B. XV
C. $U^T XV$
D. $XU^T \Sigma V$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Consider the matrix

$$A = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 2 & 0 \\ 3 & 0 & 1 \end{pmatrix}$$

Which statement is true?

A. A has three distinct eigenvalues
B. A has an eigenvalue of multiplicity two, and another eigenvalue of multiplicity one
C. A has an eigenvalue of multiplicity three
D. A has one eigenvalue of multiplicity one, and no other eigenvalues

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let P_V and P_W be the projection matrices onto the subspaces V and W, respectively. Consider the operation where a vector $\mathbf{b}$ is first projected onto the subspace W using P_W, and then the result is projected onto V using P_V. This operation is represented as:

$$\mathbf{b} \rightsquigarrow P_W\mathbf{b} \rightsquigarrow P_V P_W \mathbf{b}.$$

What happens to $P_V P_W \mathbf{b}$ if V and W are orthogonal subspaces?

A. $P_V P_W \mathbf{b} = \mathbf{b}$
B. $P_V P_W \mathbf{b} = P_V \mathbf{b}$
C. $P_V P_W \mathbf{b} = 0$
D. $P_V P_W \mathbf{b} = P_W \mathbf{b}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

For the system of linear equations $\mathbf{Ax} = \mathbf{b}$, where

$$\mathbf{A} = \begin{pmatrix} 1 & 3 \\ 1 & 4 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 3 \\ 6 \\ \beta \end{pmatrix}$$

for what value of β will the system $\mathbf{Ax} = \mathbf{b}$ have a solution?

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

Consider the following matrix and its singular value decomposition $A = U\Sigma V^T$:

$$A = \begin{pmatrix} 1/\sqrt{10} & 1/\sqrt{15} & 1/\sqrt{2} & -1/\sqrt{3} \\ -2/\sqrt{10} & 3/\sqrt{15} & 0 & 0 \\ 2/\sqrt{10} & 2/\sqrt{15} & 0 & 1/\sqrt{3} \\ -1/\sqrt{10} & -1/\sqrt{15} & 1/\sqrt{2} & 1/\sqrt{3} \end{pmatrix} \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1/\sqrt{3} & -1/\sqrt{6} & -1/\sqrt{2} \\ -1/\sqrt{3} & -2/\sqrt{6} & 0 \\ 1/\sqrt{3} & -1/\sqrt{6} & 1/\sqrt{2} \end{pmatrix}^T.$$

What will be rank of the matrix?

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Numerical Type | Award: 1 | Penalty: 0 | Calculus

Let the function be defined as:

$$f(x, y, z, w) = x^2 - y^2 + y^3 + yz + z^2 + w^2$$

. How many stationary points does $f(x, y, z, w)$ have, i.e., points for which $\nabla f = 0$?

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Find the limit.

$$\lim_{x \to 0} \frac{\sin(4x)\sin(3x)}{x^2}$$

A. 0

B. 12

C. $\frac{1}{12}$

D. $\frac{3}{4}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let X and Y be two independent binary random variables, each taking the values 0 or 1 with equal probability, i.e., each value occurs with probability $\frac{1}{2}$.

What is the probability $\mathbb{P}(X = 0 \mid X + Y = 1)$?

A. $\frac{1}{4}$

B. $\frac{1}{2}$

C. $\frac{2}{3}$

D. $\frac{3}{4}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

The random variable X has the following probability density function (pdf):

$$f(x) = \begin{cases} \frac{1}{6}x, & 2 < x \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

Determine $F(3) - F(2)$, where $F(x)$ is the cumulative distribution function of X.

A. 0

B. $\frac{5}{12}$

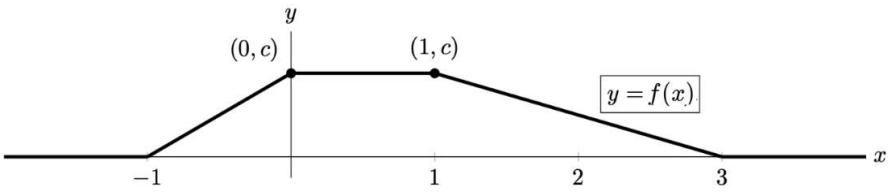
C. $\frac{9}{12}$

D. 1

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

The graph below shows a probability density function $f(x)$.



Find the value of the constant c.

A. $\frac{5}{2}$

B. $\frac{2}{5}$

C. $\frac{1}{3}$

D. $\frac{2}{3}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Three engineers, Jan, Pat, and Rory, are processing work orders. The time it takes each to complete one work order is as follows:

- Jan takes an average of 3 hours,
- Pat takes an average of 1 hour,

- Rory takes an average of 4 hours.

Due to their speeds:

- Pat processes 50% of the work orders,
- Jan and Rory each process 25% of the work orders.

What is the mean time (in hours) it takes for any given work order to be completed?

A. 2
B. $9/4$
C. $8/3$
D. 8

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Which function is not a convex function?

A. Sigmoid/Logistic function: $f(x) = 1/\left(1 + e^{-x}\right)$.
B. Linear function: $f(x) = 3x$.
C. Square function: $f(x) = x^2$.
D. ReLU function: $f(x) = \max\{x, 0\}$.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Select Type | Award: 1 | Penalty: 0 | Probability

Assume we have a sample space Ω.
Which of the following is/are TRUE?

A. If events A, B, and C are disjoint then they are independent.
B. $P(A \mid B) \propto \frac{P(A)P(B|A)}{P(A|B)}$.
C. $P(A \cup B) \geq P(A)$.
D. $P(A \cap B) \leq P(A)$.

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Suppose you train a polynomial regression model of degree $d = 3$ to approximate the quadratic function $g(x) = 7x^2 + \varepsilon$, where ε is Gaussian random variable with $\mu = 0$ and $\sigma^2 = 4$.
What is the irreducible error?

A. 2
B. 0
C. 4
D. x^3

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Select Type | Award: 1 | Penalty: 0 | Databases

Which of the following functional dependencies are in the closure of $\{T \to UV, TV \to WX, UX \to Y\}$? Mark all that apply.

A. $UX \to W$
B. $V \to X$
C. $T \to X$
D. $T \to Y$

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Numerical Type | Award: 1 | Penalty: 0 | Databases

The order of a leaf node in a B tree is the maximum number of (value, data record pointer) pairs it can hold. Given that the block size is $1\mathrm{K}$ bytes, data record pointer is 7 bytes long, the value field is 9 bytes long and a block pointer is 6 bytes long, what is the order of the leaf node?

Your Answer: | Correct Answer: 46 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Consider an unconstrained minimization problem where we aim to find a minimizer x^* of a function $f(x)$.
Statement I: A critical point x^* of f is a minimizer of f if the Hessian matrix $H_f(x^*)$ is negative definite.
Statement II: A necessary condition for f to have a minimum at x^* is that $\nabla f(x^*) = 0$.
Which of the following is correct?

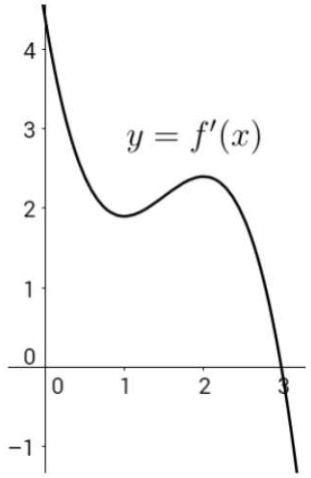
A. Both statements are true.
B. Both statements are false.
C. Statement I is true, and Statement II is false.
D. Statement I is false, and Statement II is true.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

The graph of f', the derivative of f, is pictured below:
Which of the following statements are true?

I. f has a local minimum at $x = 3$.
II. f is concave up on the interval $(1, 2)$.
III. f is increasing on the interval $(0, 3)$.



A. Only one of the statements is true.
B. I and II
C. II and III
D. I and III

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

A program selects a random integer x as follows:

- A random bit (0 or 1) is first generated uniformly, with equal probabilities.
- If the bit is 0, x is drawn uniformly from the set $\{0, 1, 2, 3\}$.
- If the bit is 1, x is drawn uniformly from the set $\{0, -1, -2, -3, -4, -5, -6, -7\}$.

What is the probability that $|x| = 1$?

A. $\frac{1}{8}$

B. $\frac{3}{16}$

C. $\frac{1}{4}$

D. $\frac{5}{16}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Numerical Type | Award: 2 | Penalty: 0 | Probability

The distance between two binary strings is the number of positions in which they differ. For example, the distance between "101" and "011" is 2.

Suppose we randomly pick two binary strings c_1 and c_2 of length 100, independently. Each position in the strings has a 0 or 1 with equal probability.

Let X_i be an indicator random variable for each position i, such that:

- $X_i = 1$ if c_1 and c_2 differ at position i.

- $X_i = 0$ otherwise.

The distance $d(c_1, c_2)$ is given by:

$$d(c_1, c_2) = \sum_{i=1}^{100} X_i$$

What is the expected value of the distance between c_1 and c_2 i.e. $\mathrm{E}[\mathrm{d}(\mathrm{c}_1, \mathrm{c}_2)]$?

Your Answer: | Correct Answer: 50 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let $Y \sim \text{Bernoulli}(1/3)$, and define X conditional on Y as follows:

- If $Y = 0$, then $X \sim \text{Bernoulli}(3/4)$.
- If $Y = 1$, then $X \sim \text{Bernoulli}(4/5)$.

Which of the following tables correctly represents the joint pmf of X and Y, with X-values in the rows and Y -values in the columns?

A.

	$Y=0$	$Y=1$
$X=0$	$1/12$	$1/6$
$X=1$	$1/3$	$2/5$

B.

	$Y=0$	$Y=1$
$X=0$	$1/6$	$1/15$
$X=1$	$1/2$	$4/15$

C.

	$Y=0$	$Y=1$
$X=0$	$1/6$	$1/6$
$X=1$	$1/3$	$1/3$

D.

	$Y=0$	$Y=1$
$X=0$	$1/4$	$1/5$
$X=1$	$3/4$	$4/5$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #31 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Consider X as a continuous random variable with the probability density function:

$$f(x) = \begin{cases} 2x & \text{for } 0 \le x \le 1 \\ 0 & \text{elsewhere} \end{cases}$$

Which of the following statements about the expectations $\mathbb{E}[X]$ and $\mathbb{E}\left[X^2\right]$ is/are correct?

A. $\mathbb{E}[X] = \frac{1}{2}$

B. $\mathbb{E}[X] = \frac{2}{3}$

C. $\mathbb{E}\left[X^2\right] = \frac{1}{3}$

D. $\mathbb{E}\left[X^2\right] = \frac{1}{2}$

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #32 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let X_1, X_2, X_3, X_4 be pairwise uncorrelated random variables, each with mean 0 and variance 1, i.e.,

$$\mathrm{Cov}(X_i, X_j) = 0 \quad \text{for} \quad i \neq j$$

The correlation between $(X_1 + X_2)$ and $(X_2 + X_3)$ is given as:

$$\rho\left(X_1 + X_2, X_2 + X_3\right) = \frac{\mathrm{Cov}(X_1 + X_2, X_2 + X_3)}{\sqrt{\mathrm{Var}(X_1 + X_2) \cdot \mathrm{Var}(X_2 + X_3)}}$$

Compute $\rho\left(X_1 + X_2, X_2 + X_3\right)$.

A. 0

B. $\frac{1}{2}$

C. 1

D. $\frac{1}{\sqrt{2}}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #33 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Which of the following is true, given the optimal learning rate?

A. For convex loss functions (i.e. with a bowl shape), batch gradient descent is guaranteed to eventually converge to the global optimum while stochastic gradient descent is not.
B. For convex loss functions (i.e. with a bowl shape), stochastic gradient descent is guaranteed to eventually converge to the global optimum while batch gradient descent is not.
C. For convex loss functions (i.e. with a bowl shape), both stochastic gradient descent and batch gradient descent will eventually converge to the global optimum.
D. For convex loss functions (i.e. with a bowl shape), neither stochastic gradient descent nor batch gradient descent are guaranteed to converge to the global optimum.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #34 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Alice has collected a dataset of dependent and independent variables $\{(x_1, y_1), \ldots, (x_n, y_n)\}$ and performs linear regression on it, obtaining the weight vector w_{Alice}. Bob collects the same dataset but records each feature in different units. Below is a toy dataset with 2 features and 3 examples, as shown in the tables:

a	p
b	q
c	r

Alice Table

2a	3p
2b	3q
2c	3r

Bob Table

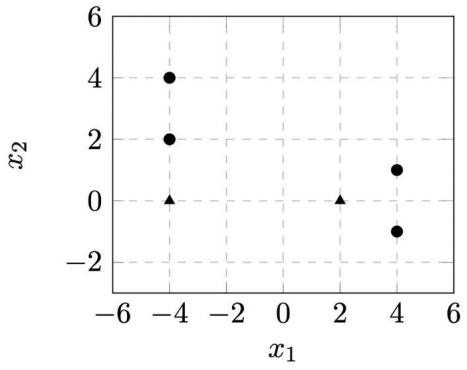
Which of the following statements is correct about the training loss and weight vectors w_{Alice} and w_{Bob}, obtained using linear regression without regularization?

A. Alice and Bob have the same training loss, and $w_{\text{Alice}} = w_{\text{Bob}}$.
B. Alice and Bob have different training losses, and $w_{\text{Alice}} \neq w_{\text{Bob}}$.
C. Alice and Bob have the same training loss, but $w_{\text{Alice}} \neq w_{\text{Bob}}$.
D. Alice and Bob have different training losses, but $w_{\text{Alice}} = w_{\text{Bob}}$.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #35 | Numerical Type | Award: 2 | Penalty: 0 | Machine Learning

A company named Amazing, Inc. is using k-means clustering to determine the optimal locations for their k stores in a given region.
Amazing, Inc. has four customers in Megacity, located at positions (x1, x2), represented by ● (circles) in the plot below. They plan to build two stores, with initial guesses for store locations represented by ▲ (triangles) in the plot.



Using the k-means loss function:

$$L(\mu) = \sum_{i=1}^{n} \min_{j=1,\ldots,k} \left\| x^{(i)} - \mu^{(j)} \right\|^2$$

What is the final loss $L(\mu)$ after the k-means algorithm converges and the stores (triangles) are optimally placed?

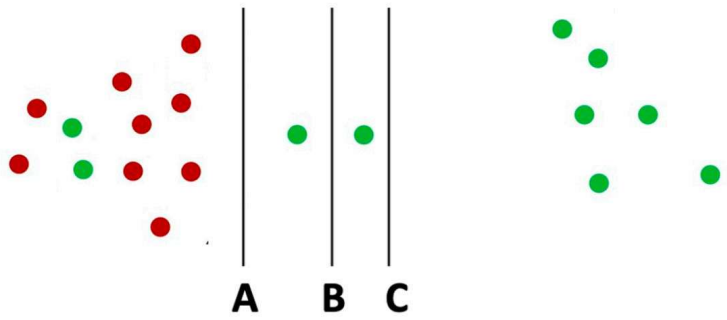
Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #36 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider a Soft SVM applied to a training set three times, each with a different value of the hyperparameter C (the slack penalty weight). The results of these runs are as follows:

- In the first run, with $C = C_A$, the decision boundary is denoted as A.
- In the second run, with $C = C_B$, the decision boundary is denoted as B.
- In the third run, with $C = C_C$, the decision boundary is denoted as C.

Which of the following options correctly orders C_A, C_B, and C_C by their magnitude?



A. $C_A < C_B < C_C$

B. $C_C < C_B < C_A$

C. $C_A = C_B = C_C$

D. $C_B < C_C < C_A$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #37 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Which of the following statements about gradient descent are TRUE?

A. After a gradient descent update step, the objective function value at the weight vector is always lower after the update than before.
B. There is always a unique steepest descent direction in gradient descent.
C. Gradient descent converges to a globally optimal solution for logistic regression under appropriate assumptions.
D. Since ReLU is a convex function, a neural network that uses ReLU activations is also a convex function, and therefore gradient descent will converge to a globally optimal solution on neural networks with ReLU activations.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #38 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Suppose we have a data set which we divide into 3 equally sized parts, A, B, and C. We fit 3 linear regression models with $L2$ regularization (i.e. ridge regression), X, Y, and Z, all on A. Each model uses the same features and training set, the only difference is the λ used by each model. Select all below that are always true.

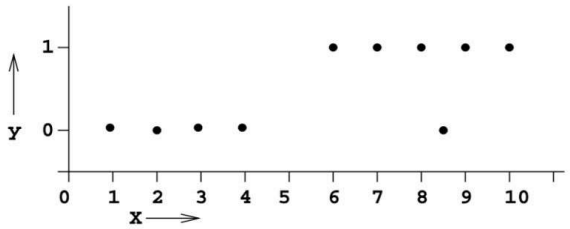
A. Suppose Z has the lowest average loss on B. Model Z will have the lowest average loss when evaluated on C.
B. If A and B have the same exact mean and variance, the average loss of model Y on B will be exactly equal to the average loss of Y on A.
C. If $\lambda = 0$ for model X, $\text{Loss}(X, A) \leq \text{Loss}(Y, A)$ and $\text{Loss}(X, A) \leq \text{Loss}(Z, A)$.
D. If $\lambda_Y < \lambda_Z$, then $\text{Loss}(Y, A) \leq \text{Loss}(Z, A)$.

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #39 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Assume we will learn a decision tree from this data.

X	Y
1	0
2	0
3	0
4	0
6	1
7	1
8	1
8.5	0
9	1
10	1



Assume that when the decision tree splits on the real valued attribute x, it puts the split threshold halfway between the attributes that surround the split. For example, using information gain as the splitting criterion, the decision tree would initially choose to split at $x = 5$, which is halfway between the $x = 4$ and $x = 6$ data points.
Let Algorithm DT2 be the method of learning a decision tree with only two leaf nodes (i.e. only one split). What will be the training error and leave-one-out cross-validation (LOOCV) error of DT2?

A. Training error $= 1/10$, LOOCV error $= 1/10$
B. Training error $= 0/10$, LOOCV error $= 1/10$
C. Training error $= 1/10$, LOOCV error $= 2/10$
D. Training error $= 0/10$, LOOCV error $= 2/10$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #40 | Multiple Select Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Consider performing tree search for some search graph.

Let depth (n) be the depth of search node n and $\text{cost}(n)$ be the total cost from the start state to node n. Let G_d be a goal node with minimum depth, and G_c be a goal node with minimum total cost.

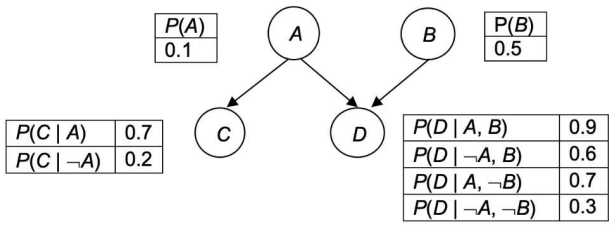
For iterative deepening (where we repeatedly run DFS and increase the maximum depth allowed by 1), mark all conditions that are guaranteed to be true for every node n that could be expanded during the search.

A. $\text{cost}(n) \leq \text{cost}(G_c)$
B. $\text{cost}(n) \leq \text{cost}(G_d)$
C. $\text{depth}(n) \leq \text{depth}(G_c)$
D. $\text{depth}(n) \leq \text{depth}(G_d)$

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #41 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

Consider the following Bayesian Network containing four Boolean random variables.



What is $P(A \mid B, C, D)$?

A. 0.0315
B. 0.0855
C. 0.368
D. 0.583

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #42 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider the following statements regarding pre- and post-visit labels in depth-first search (DFS):
pre is start time and post is finish time.

S1: In an undirected graph, if the DFS pre- and post-visit labels of two vertices u and v satisfy the condition: pre(u) < post(u) < pre(v) < post(v), then no edge can exist between u and v.

S2: In a directed graph, if the DFS pre- and post-visit labels of two vertices u and v satisfy the condition: pre(u) < post(u) < pre(v) < post(v), then no edge (in either direction) can exist between u and v.

Which of the following is correct?

A. Both S1 and S2 are true.
B. Only S 1 is true.
C. Only S2 is true.
D. Neither S1 nor S2 is true.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #43 | Multiple Select Type | Award: 2 | Penalty: 0 | Data Structures

Consider a complete binary tree (where all levels are fully filled except possibly for the last level, which is filled from left to right).

Which of the following single tree traversals is sufficient to uniquely reconstruct the tree, assuming each node has a distinct value?

A. Preorder
B. Inorder
C. Postorder
D. None of the above - No single traversal can uniquely reconstruct the tree

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #44 | Numerical Type | Award: 2 | Penalty: 0 | Data Structures

Consider a rooted binary tree with distinct node values, where the tree has a height 10. Note: A tree with only the root has height 0. The tree has the property that its in-order and post-order traversals are same. What is the maximum number of nodes that can be present in such a tree?

Your Answer: | Correct Answer: 11 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #45 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in Python

Given the list $[1, 2, 3, 7, 2, 4]$ as input, the following function separate is executed:

```
    def separate(data):
        if len(data) == 0:
            return []
        # odds
5.      if data[0] % 2 != 0:
            return data[0:1] + separate(data[1:])
        # evens
        else:
            return separate(data[1:]) + data[0:1]
```

Which of the following could be the output of the function?

A. $[1, 3, 7, 2, 4, 2]$
B. $[1, 3, 7, 4, 2, 2]$
C. $[1, 7, 3, 2, 2, 4]$
D. $[3, 7, 1, 2, 2, 4]$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #46 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let A be a 4×4 matrix with the characteristic equation $\lambda^4 - 3\lambda^2 = 0$, such that A is not diagonalizable. Which of the following statements is true?

A. Rank of A is 1
B. Rank of A is 2
C. Rank of A is 3
D. It is impossible to determine the rank of A from the given information

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #47 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Consider the quadratic form $f(x, y, z) = 3x^2 + 8xy + 2xz + 6y^2 + 4yz$.

Which statement is true?

A. f is positive definite
B. f is positive semi-definite but not positive definite
C. f is negative semi-definite but not negative definite
D. f is indefinite

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #48 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Suppose that

$$\left[\begin{bmatrix}1\\2\end{bmatrix}\right]_{\mathcal{B}} = \begin{bmatrix}3\\4\end{bmatrix} \text{ and } \left[\begin{bmatrix}3\\5\end{bmatrix}\right]_{\mathcal{B}} = \begin{bmatrix}2\\3\end{bmatrix}.$$

Which of the following is the basis $\mathcal{B} = \{v_1, v_2\}$ for $\mathbb{R}^2$? A. $\left\{\begin{bmatrix} -9 \\ -14 \end{bmatrix}, \begin{bmatrix} 7 \\ 11 \end{bmatrix}\right\}$ B. $\left\{\begin{bmatrix} -11 \\ -14 \end{bmatrix}, \begin{bmatrix} 7 \\ 9 \end{bmatrix}\right\}$ C. $\left\{\begin{bmatrix} -9 \\ 14 \end{bmatrix}, \begin{bmatrix} 7 \\ -11 \end{bmatrix}\right\}$ D. $\left\{\begin{bmatrix} -11 \\ -14 \end{bmatrix}, \begin{bmatrix} -7 \\ 9 \end{bmatrix}\right\}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #49 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let A be a block matrix with each block of size 2×2:

$$A = \left[\begin{array}{cc|cc} 2 & 1 & 3\alpha & 4 \\ 0 & \alpha - 1 & 4 & 0 \\ \hline 0 & 0 & 2 & 1 \\ 0 & 0 & \alpha & 4 \end{array}\right]$$

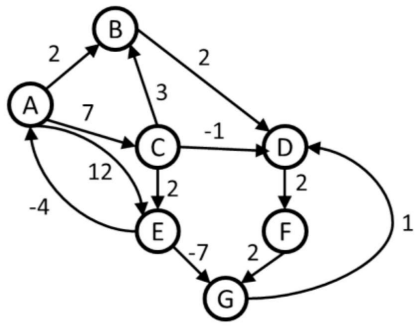
Find the values of α for which A is singular.

A. $\alpha = 0$ and $\alpha = 4$
B. $\alpha = 2$ and $\alpha = 3$
C. $\alpha = 1$ and $\alpha = 8$
D. $\alpha = 0$ and $\alpha = 1$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #50 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Consider the following directed, weighted graph:



Even though the graph has negative weight edges, we use Dijkstra's algorithm to calculate supposedly shortest paths from A to every other vertex.
Which of the following is/are correct?

A. The order in which vertices are added to the shortest path tree using Dijkstra's algorithm is : A B D F G C E.
B. The order in which vertices are added to the shortest path tree using Dijkstra's algorithm is : A B D F C G E.
C. Dijkstra's algorithm finds the wrong path to some of the vertices.
D. Dijkstra's algorithm finds the right path to all of the vertices.

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #51 | Numerical Type | Award: 2 | Penalty: 0 | Databases

Consider the relation defined by this statement:

```
CREATE TABLE Students(id integer, gpa float,
name text, address text, gender char,
PRIMARY KEY (id));
```

Consider the following query and the table of data to the right:

id	gpa	name	address	gender
123	null	Joe	38 Maple	M
124	3.2	Hui	64 Vine	F
127	3.9	Celia	21 Elm	F
111	3.2	Hector	11 Oak	M

The SQL query is as follows:

```
SELECT S.id
FROM Students S
WHERE S.gpa > 3.3
    AND S.id > 120;
```

How many rows should be in the output?

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #52 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the attribute set $R = ABCDEF$ and the functional dependency set

$$F = \{C \rightarrow D, A \rightarrow B, \mathrm{B} \rightarrow \mathrm{EF}, \mathrm{F} \rightarrow \mathrm{A}\}$$

.

A GO-Classes student decides to write a decomposition of tables as ABDE and ABCF. Right now, this is not a lossless decomposition. Which of the following changes (applied individually, not combined together) would make this a lossless decomposition?

A. Adding $\mathrm{D} \rightarrow \mathrm{ABCDEF}$ to the functional dependency set F
B. Adding $\mathrm{E} \rightarrow \mathrm{C}$ to the functional dependency set F
C. Adding $\mathrm{B} \rightarrow \mathrm{D}$ to the functional dependency set F

A. All A, B, C

B. Only B

C. Only C

D. Only B and C

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #53 | Multiple Select Type | Award: 2 | Penalty: 0 | Data Structures

The following is a partial level-order traversal of a binary search tree (BST):

$$80 \quad 40 \quad 110 \quad ? \quad 100 \quad 120 \quad 60$$

Which of the keys below could be the one labeled with a question mark?

A. 50

B. 70
C. 90
D. 115

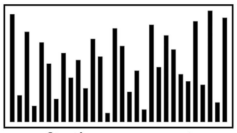
Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #54 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Suppose we want to sort a collection of sticks according to height.

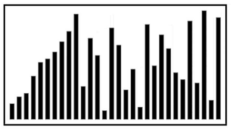
We have observed the intermediate result shown in the image, where sticks are partially sorted during the process.

From:



Starting arrangement

To:



Intermediate Result

Based on this observation, which sorting algorithm are we most likely applying?

A. Bubble Sort
B. Insertion Sort
C. Selection Sort
D. Merge Sort

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #55 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Which of the following options are enough evidence to prove or disprove if a given comparison-based sorting algorithm works correctly?

A. Professor "A" gives a comparison-based sorting algorithm that runs in $\mathrm{T}(n) = 2\mathrm{T}(n-1) + 1$. This is sufficient information to conclude the correctness/incorrectness of A's algorithm.
B. Professor "B" gives a comparison-based sorting algorithm that runs in $\mathrm{T}(n) = 4\mathrm{T}(n/5) + 5n \log \log n$. This is sufficient information to conclude the correctness/incorrectness of B's algorithm.
C. Professor "C" gives a comparison-based sorting algorithm that runs in $\mathrm{T}(n) = 7\ \mathrm{T}(n/6) + n/42$. This is sufficient information to conclude the correctness/incorrectness of C's algorithm.
D. Professor "D" gives a comparison-based sorting algorithm that runs in $\mathrm{T}(n) = \mathrm{T}(n-1) + 1$. This is sufficient information to conclude the correctness/incorrectness of D's algorithm.

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

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